

NEUROPHARMACOLOGY LAB REPORT — LONGITUDINAL STUDY EXPERIMENT REPORT

TARGET: ORGANISM TYPE: LONGITUDINAL

COMPOUND DETAILS

Compound	Methylphenidate	Drug Class	NDRI
Dose (mg eq.)	20mg	BBB Coefficient	0.75
Receptor Targets	DAT, NET, D1, D2, NMDA, AMPA		
Session Date	2026-06-16T22:38:57.357Z		

Tests run in an isolated Controlled Lab session. Live brain was not modified. Platform biologically constrained by HCP-MMP1.0, Allen Brain Atlas, and Brainnetome Atlas data.

PEAK RECEPTOR OCCUPANCY

DAT	NET	D1	D2	NMDA	AMPA
4.0%	4.0%	4.0%	4.0%	4.0%	4.0%

Occupancy curves follow rapid absorption / first-order elimination kinetics (high-level approximation). Not equivalent to full pharmacokinetic modeling.

REGIONAL ACTIVATION EFFECTS

Region	Receptor	Activation Delta	Predicted Effect (consistent with literature)
Thalamus	D2	-21.2%	increased compulsive motor patterns
AnteriorCingulateCortex	NMDA	-20.9%	impaired memory encoding
MotorCortex	AMPA	-20.8%	network hyperexcitability
SensoryCortex	AMPA	-20.6%	network hyperexcitability
NucleusAccumbens	D1	-20.5%	impaired prefrontal gating
BasalGanglia	D1	-20.3%	impaired prefrontal gating
NucleusAccumbens	D2	-20.2%	increased compulsive motor patterns
PrefrontalCortex	NMDA	-20.1%	impaired memory encoding
PrefrontalCortex	D1	-20.0%	impaired prefrontal gating
Hippocampus	AMPA	-19.9%	network hyperexcitability
BasalGanglia	D2	-19.8%	increased compulsive motor patterns
Hippocampus	NMDA	-19.7%	impaired memory encoding

Effects represent simulated predictions from population-level receptor binding models. Not equivalent to in vivo pharmacology data.

VALIDATION SUMMARY

Integrity: PASS Stability: PASS Verdict: WEAK EFFECT

ANSInteroceptiveCoupling did not produce a measurable separation from baseline or shuffled control on pre-registered metrics. Claims of biological significance are not supported by this

run.

Language note: This report uses "consistent with", "suggests", or "did not support" for all mechanistic claims. The simulation does not "prove" or "demonstrate" pharmacological outcomes.

PHARMACOKINETIC PROFILE

Cmax (normalized)	0.054	Tmax	28.7 ticks
Half-life	33 ticks	ke (elimination)	0.021
AUC	2.58	EC50	0.342
Hill Coefficient	1.6	BBB Penetration	0.75

PK parameters derived from two-compartment rapid-absorption / first-order elimination model. Not equivalent to full clinical PK study.

NEUROCHEMICAL CASCADE MAP

Neurochemical	Net Delta
Dopamine	+9.0%
BDNF	+5.0%
Adenosine	-1.0%
serotonin	+3.0%
NPY	-2.0%
Glutamate	+11.0%
Nitric Oxide	-2.0%
Calcium	+3.0%

From	To	Mechanism	Delta
Dopamine	BDNF	agonist cascade	+5.0%
BDNF	Adenosine	agonist cascade	+4.0%
Dopamine	serotonin	agonist cascade	+4.0%
serotonin	NPY	agonist cascade	+3.0%
Glutamate	BDNF	agonist cascade	+5.0%
BDNF	Nitric Oxide	agonist cascade	+3.0%
Glutamate	Calcium	agonist cascade	+5.0%
Calcium	BDNF	agonist cascade	+3.0%

Neurochemical deltas are derived from receptor-binding-to-second-messenger cascade models. Cascade sequence represents dominant literature pathway, not exhaustive pharmacology.

TIME-SERIES RESPONSE

Tick	Time (ms)	Plasma Concentration
0	0	0.0000
5	4,365	0.0010
10	8,730	0.0030
15	13,095	0.0040
20	17,460	0.0050

25	21,825	0.0070
30	26,190	0.0080
35	30,555	0.0090
40	34,920	0.0100
45	39,285	0.0120
- ... -		
575	501,975	0.0330
580	506,340	0.0330
585	510,705	0.0330
590	515,070	0.0330
595	519,435	0.0320

First 10 + last 5 time points shown. Full series: 120 ticks at 4365ms/tick intervals.

STATISTICAL VALIDATION

Cohen's d	0.18	Effect Size Class	small
95% CI Lower	-0.219	95% CI Upper	0.369
ESURIENS Hunger-Drive Impact	13.4%	Memory Temple Encoding Δ	46.3%

Statistical parameters estimated from single-session simulation. Cohen's d < 0.2: small, 0.2-0.5: medium, 0.5-0.8: large, > 0.8: very large. Confidence interval assumes gaussian error propagation.